

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation function $f(\cdot)$ and parameter matrices R, W, V . Let θ denote the parameters shared in time and let $\theta(t)$ denote their usage at time t . Further, let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial \theta}$ is a column vector.

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Verbergen

Which statements about RTRL are true?

- ☐ a. The term $\frac{\partial s(t)}{\partial \theta}$ has $O(N^4)$ elements, where N denotes the number of hidden units.
- ☐ b. RTRL computes the gradients $\frac{\partial s(t)}{\partial \theta}$ during the forward pass already while BPTT has to go through the sequence twice (forward and backward).
- ☐ c. BPTT considers the recursion $\frac{\partial L}{\partial s(t)} = \frac{\partial s(t+1)}{\partial s(t)} \frac{\partial L}{\partial s(t+1)} + \frac{\partial L(t)}{\partial s(t)}$.
- ☐ d. Schmidhuber's approach divides the input sequence into chunks of size N (number hidden units) and performs RTRL within these chunks. Then it uses BPTT to consolidate the gradients for these chunks.
- ☐ e. RTRL considers the recursion $\frac{\partial s(t)}{\partial \theta} = \frac{\partial s(t)}{\partial \theta(t)} + \frac{\partial s(t-1)}{\partial \theta} \frac{\partial s(t)}{\partial s(t-1)}$.



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Which statements about LSTM dropout are correct?

- ☐ a. It is especially important to apply dropout when evaluating the model.
- ☐ b. The central idea of zoneout (Krueger et al., 2017) is to randomly clear the inputs $x(t)$ instead of the memory cell $c(t)$.
- ☐ c. LSTM needs special dropout techniques because with a naive application the probability of a memory cell to "survive" decreases exponentially as the sequence grows in time.
- ☐ d. LSTM needs special dropout techniques because a naive application of dropout will result in vanishing/exploding gradients.
- ☐ e. Dropout is a regularization technique that randomly sets activations to zero during the forward pass.

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Verbergen

Which of the following statements are true regarding the Mamba architecture?

- ☐ a. Mamba is considered a selective State Space Model (SSM)
- ☐ b. Mamba's memory requirements scale quadratically $\mathcal{O}(n^2)$ with input sequence length.
- ☐ c. The Mamba architecture combines the H3 block of SSM with the MLP block structure of transformer.
- ☐ d. Mamba's selective SSM differs from traditional SSMs because its (**A**, **B**, **C**) functions are independent of the input.
- ☐ e. Mamba uses pre-activation forget (sd), input (sb), and output (sc) gates - these gates control the flow of information within the Mamba block.

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The Elman Network (without output activation function and bias units) can be defined as

$$s(t) = Wx(t) + a(t-1)$$

$$a(t) = f(s(t))$$

$$\hat{y}(t) = Va(t)$$

with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$, and parameter matrices W , V . Which of the following statements are true?

- ☐ a. The hidden units of the Elman network are disconnected in the sense that they cannot exchange information from one time step to another.
- ☐ b. The recurrent Jacobian of the hidden activations is $\frac{\partial a(t)}{\partial a(t-1)} = \text{diag}(f'(s(t-1)))$.
- ☐ c. The recurrent weights are equal to the identity matrix.
- ☐ d. Elman proposed to train this network locally in time, i.e., on each time step separately.
- ☐ e. The recurrent weights are variable.

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Verbergen

Which statements about LSTM variants are true?

- ☐ a. The key idea of multidimensional LSTM (Graves et al., 2007) is to enable the network to process sequences consisting of vectors which stem from high-dimensional vector spaces.
- ☐ b. Convolutional LSTM (Shi et al., 2015) can process sequences of images, e.g., videos.
- ☐ c. Stacked LSTM (Graves et al., 2013) stack LSTM layer by using one LSTM's final cell state as the initial cell state for the next LSTM.
- ☐ d. The Bidirectional LSTM (Schuster et al., 1997) has two layers. The second layer receives input from the first layer and reverses the processing direction.
- ☐ e. Peephole connections (Gers and Schmidhuber, 2000) provide gates with direct read access to the cell state.

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Verbergen

Which statements about LSTM are true?

- ☐ a. To counter the drift effect, the input gate should be initialized with positive values.
- ☐ b. "Ticker steps" refers to the technique of letting an LSTM run for more time steps even after the input sequence has been taken in by the network.
- ☐ c. An LSTM with coupled input gate and forget gate, i.e.,
$$c(t) = (1 - i(t)) \odot c(t-1) + i(t) \odot z(t),$$
 leads (alongside other changes) to the gated recurrent unit (GRU).
- ☐ d. It is often beneficial to initialize the forget gate with positive bias values.
- ☐ e. Most LSTM networks used in practice do not feature a forget gate.
- ☐ f. The forget gate reintroduces the vanishing gradient problem to a certain degree.
- ☐ g. The focused LSTM network feeds the input vector $x(t)$ only into input gate $i(t)$ and output gate $o(t)$ but not into the block input $z(t)$.
- ☐ h. The inputs to the lightweight LSTM have to fulfill the Markov property.

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Which of the following statements best describes the role of **exponential gating** in xLSTM?

- ☐ a. It addresses the vanishing gradient problem by scaling down older inputs and enabling the network to revise storage decisions for nearest neighbor search.
- ☐ b. It primarily controls the flow of information through the output gate.
- ☐ c. It is used to introduce randomness into the training process, preventing overfitting.
- ☐ d. It introduces a linear transformation to the network's hidden state.
- ☐ e. It primarily increases the computational complexity of xLSTM by introducing linearities

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Which of the following statements is true regarding the memory of mLSTM?

- ☐ a. The size of the memory matrix is constant and depends on W_k .
- ☐ b. The size of the memory matrix $\mathbf{C} \in \mathbb{R}^{d \times d}$ is dependent on the sequence length L .
- ☐ c. The size of the matrix memory of mLSTM grows linearly with the length of the input sequence L .
- ☐ d. The memory matrix $\mathbf{C} \in \mathbb{R}^{d \times d}$ is independent of sequence length L and has a capacity dependent on d .
- ☐ e. The matrix memory is computed by taking the old matrix memory and adding a covariance update rule using $v_t k_t^T$.

$$\hat{y}(t) = Va(t)$$

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with input vectors $x(t)$, hidden pre-activation vectors $s(t)$, hidden activation vectors $a(t)$, activation function $f(\cdot)$ and parameter matrices R, W, V . Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss function at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\delta(t) = \frac{\partial L}{\partial s(t)}$ is a column vector. Which of the following statements are true?

- ☐ a. The asymptotic complexity of BPTT is $O(T^2)$.
- ☐ b. The gradient of the loss with respect to the input weights W can be written as $\frac{\partial L}{\partial W} = \sum_{t=1}^T \delta(t)x^\top(t)$.
- ☐ c. The gradient of the loss with respect to the recurrent weights R can be written as $\frac{\partial L}{\partial R} = \sum_{t=1}^T \delta(t)a^\top(t-1)$.
- ☐ d. BPTT is a common regularization technique for recurrent neural networks.
- ☐ e. The deltas fulfill the recursive relation $\delta(t) = \text{diag}(f'(s(t))) \left(V^\top \frac{\partial L(t)}{\partial \hat{y}(t)} + R^\top \delta(t-1) \right)$.

denote the loss at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-laying convention, i.e., $\frac{\partial L}{\partial c(t)}$ is a column vector. The diag operator turns a vector into a diagonal matrix, i.e., $\text{diag}((1, 1)^\top) = I \in \mathbb{R}^{2 \times 2}$ and $\odot$ denotes Hadamard's product. Which of the following statements are true?

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- ☐ a. If we choose $z(t) = \sigma(Ux(t) + Ph(t-1))$, then $E[z(t)] > 0$ and the memory cells will always increase at every time step. This can be a problem for very long sequences but can also be helpful for certain problems.
- ☐ b. The gradient of the loss with respect to the hiddens is $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)} + U \text{diag}(1 - z(t+1)^2) \frac{\partial L}{\partial z(t+1)} + V \text{diag}(i(t+1)(1 - i(t+1))) \frac{\partial L}{\partial i(t+1)}$
- ☐ c. Because of the simple structure of the memory cell, the LSTM architecture fails to be Turing complete.
- ☐ d. The LSTM architecture has no exploding or vanishing gradients because $\frac{\partial h(t)}{\partial h(t-1)}$ is an orthogonal matrix.
- ☐ e. The memory cell fulfills $\frac{\partial c(t)}{\partial c(t-\tau)} = I$ (neglecting dependencies via the hiddens) for any $\tau \in \{1, \dots, t-1\}$, where I is the identity matrix. This solves the vanishing gradient problem and is called constant error carousel.



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Verbergen

Backpropagation through time (BPTT) is the standard gradient computation algorithm for RNNs. However, there exist alternatives, e.g., truncated BPTT and real-time recurrent learning (RTRL). Which statements about these algorithms are true.

- ☐ a. Truncated BPTT gradients are different from BPTT gradients.
- ☐ b. Truncated BPTT has advantages over BPTT if the input sequences are short.
- ☐ c. RTRL is very fast and therefore often used in practice.
- ☐ d. Truncated BPTT divides the input sequence into chunks and applies BPTT within these chunks but not across them.
- ☐ e. The memory complexity of RTRL is $O(N^3)$, where N denotes the number of hidden units.
- ☐ f. RTRL gradients are different from BPTT gradients.

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Which statements about attention mechanism are true?

- ☐ a. An LSTM with self-attention has access to all previous hidden states and cell states via a softmax-based attention mechanism.
- ☐ b. The attention score $e(i)$ is obtained from the attention vector $a(i)$ via the softmax function.
- ☐ c. The scaled dot-product attention mechanism generally has the form $\text{softmax}\left(\frac{1}{\sqrt{d}} QK^\top\right)V$, where $K, Q, V \in \mathbb{R}^{L \times d}$ are keys, queries, and values, respectively, and L is the sequence length and d is the hidden dimension.
- ☐ d. The LSTM gates can be interpreted as a temporal attention mechanism.
- ☐ e. A spatial attention mechanism causes the network to treat each pixel in an image as equally important.

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Which statements about Transformer networks (Vaswani et al., 2017) are true?

- ☐ a. The Transformer architecture consists of an encoder and a decoder part, where the encoder depends on the output of the decoder.
- ☐ b. BERT (Devlin et al., 2019) uses two different pre-training objectives, (1) inserting missing (masked out) words into a sentence, and (2) next sentence prediction, i.e., generating a response sentence to a question sentence.
- ☐ c. Transformer networks use multi-head attention, which works as follows. We run multiple instances of scaled dot-product attention, concatenate their outputs (or context) and feed the result into a linear layer.
- ☐ d. The Transformer architecture suffers from the vanishing gradient problem in the time dimension.
- ☐ e. Without positional embeddings and without causal attention mask, the Transformer network has no way of figuring out the order of elements in an input sequence.

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Verbergen

Which of the following statements about RNNs are true?

- ☐ a. Many RNN architectures are Turing complete. This means that they can be trained to exactly imitate the behavior of any computer program.
- ☐ b. RNNs can be applied to input sequences of varying lengths.
- ☐ c. An RNN is equivalent to a feedforward network applied to the input and an internal state representation that depends on previous time steps.
- ☐ d. The number of RNN parameters grows linearly with the sequence length.
- ☐ e. An RNN is equivalent to a feedforward network applied to the input at every time step in parallel.

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Which statements about the self-attention mechanism are correct?

- ☐ a. In the standard self-attention formulation, cosine similarity is used as a measure of alignment between keys and values.
- ☐ b. The raw attention scores are normalized by applying a sigmoid activation function row-wise to the attention matrix.
- ☐ c. Similar to the hidden state in an LSTM, the attention matrix is updated in a recurrent manner for each token in an input sequence.
- ☐ d. The Multi-head Attention mechanism consists of a total of four weight matrices.
- ☐ e. The vector at position t in the output sequence of a self-attention layer is computed by summing the value vectors, each weighted by its corresponding attention score from the t -th row of the attention matrix.

$$h(t) = \tanh(c(t)) \odot o(t)$$

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with input vectors $x(t)$, hidden activation vectors $h(t)$, memory cell state vectors $c(t)$, gate activation vectors $z(t), i(t), o(t)$, weight matrices P, Q, R, U, V, W . Let $L(t) = L(y(t), \hat{y}(t))$ denote the loss at time t and let $L = \sum_{t=1}^T L(t)$ denote the total loss. We use denominator-layout convention, i.e., $\frac{\partial L}{\partial c(t)}$ is a column vector. The diag operator turns a vector into a diagonal matrix, i.e., $\text{diag}((1, 1)^\top) = I \in \mathbb{R}^{2 \times 2}$ and $\odot$ denotes Hadamard's product. Which of the following statements are true?

- ☐ a. Since LSTM has no vanishing gradients, stacking many LSTM layers on top of each other became common practice.
- ☐ b. Changing the definition of the cell state to $c(t) = z(t) \odot i(t)$ reintroduces the vanishing gradient problem.
- ☐ c. The gradient of the loss with respect to the hiddens was originally proposed to be approximated by $\frac{\partial L}{\partial h(t)} = \frac{\partial L(t)}{\partial h(t)}$.
- ☐ d. The gradient of the loss with respect to the memory cell is $\frac{\partial L}{\partial c(t)} = \frac{\partial L}{\partial c(t-1)} + o(t) \odot \tanh'(c(t)) \odot \frac{\partial L}{\partial h(t)}$.



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Which statements about the gated recurrent unit (GRU, Cho et al., 2014) are true?

- ☐ a. The so-called "update gate" of the GRU acts as a coupled input and forget gate.
- ☐ b. In the GRU, a so-called "reset gate" $r(t)$ regulates how strongly the increments $z(t)$ depend on the old hidden states $h(t-1)$.
- ☐ c. The GRU has less parameters than an LSTM without forget gate that has the same number of hidden units.
- ☐ d. The GRU has one gate more than the fully featured LSTM architecture.
- ☐ e. A forget gate mechanism is the core of the GRU architecture.